

**A  
Project Report  
on  
Industrial Component Inventory  
Analytics  
Developed By  
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developed at  
**Vermin Industries Pvt. Ltd -  
Vadodara**  
as  
**Partial Fulfillment of IV<sup>th</sup> Semester of  
Master of Computer Applications  
for A.Y. 2025 - 2026****

**Under The Guidance of**

**Prof. Vijya Tulsani**

**Submitted To**

**Department of MCA  
Faculty of IT & Computer Science  
PARUL University**





## **CERTIFICATE**

This is to certify that **Mr. Tushar Yadav, Harshit Uppal** Enrollment No. **2405112120198** and **2405112130039** student of Master of Computer Applications has satisfactorily completed the Major Project on “**Industrial Component Inventory Analytics**” at “**Vermin Industries Pvt. Ltd**” as fulfilment of MCA Semester IV.

Seat No. \_\_\_\_\_

Date of Submission: \_\_\_\_\_

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Internal Guide

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Project Coordinator

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Director - MCA

**Department of MCA**  
**Faculty of IT & Computer Science**  
**PARUL University, Vadodara**

**Vermin Industry Pvt. Ltd.**

# **INTERNSHIP COMPLETION CERTIFICATE**

*This is to certify that*

**Tushar Yadav**

This is to certify that **Mr. Tushar Yadav** (Enrollment No.: **2405112120198**), a student of **Master of Computer Applications (MCA)** at **Parul University**, has successfully completed his internship in the domain of **Junior Data Analyst** at our organization, **Vermin Industry Pvt. Ltd.** from **1st December 2025 to 30th March 2026**.

During this period, he demonstrated excellent analytical and technical skills, showing dedication, responsibility, and precision in data analysis, reporting, and dashboard development tasks.

*We wish him great success in all his future endeavors.*



**Gaurang Patel**

Director.

**Vermin Industry Pvt. Ltd.**

# INTERNSHIP COMPLETION CERTIFICATE

*This is to certify that*

**Harshit Uppal**

This is to certify that **Mr. Harshit Uppal** (Enrollment No.: **2405112130039**), a student of **Master of Computer Applications (MCA)** at **Parul University**, has successfully completed his internship in the domain of **Junior Data Analyst** at our organization, **Vermin Industry Pvt. Ltd.** from **1st December 2025 to 30th March 2026**.

During this period, he demonstrated excellent analytical and technical skills, showing dedication, responsibility, and precision in data analysis, reporting, and dashboard development tasks.

*We wish him great success in all his future endeavors.*



**Gaurang Patel**

Director.

## **PREFACE**

In today's rapidly evolving business environment, the ability to make fast and data-driven decisions has become essential for organisations of all sizes. This project, titled Inventory Analytics System, is developed with the objective of digitising and centralising the inventory management process at Vermin Industries, Vadodara, replacing fragmented manual Excel-based workflows with a structured, data-driven platform.

The core functionality of the system revolves around managing four key entities :- Suppliers, Components, Purchase Orders, and Inventory Stock through a normalised MySQL relational database. The application is built using Python and the Streamlit framework, providing a full-featured web interface that supports complete CRUD operations, a live SQL Query Runner using in-memory SQLite, and CSV export capability making it both a data management tool and an ad-hoc analytical reporting platform.

A key highlight of this project is its 7-page interactive Microsoft Power BI dashboard, featuring 22 custom DAX measures. The dashboard provides real-time KPI reporting covering total PO value (Rs. 1.58M), delivery rate (94.17%), automated critical stock alerts (42 items), India vs Korea supplier benchmarking, and an Overall Inventory Health Score of 84% giving management clear, actionable visibility into business performance.

Overall, this project is a practical implementation of full-stack data application development combined with business intelligence reporting. It demonstrates how Python, Streamlit, MySQL, and Power BI can be integrated into a production-quality system that delivers real value. The project has been a comprehensive learning experience covering database design, web development, DAX measure writing, and the complete software development lifecycle.

## Acknowledgement

I take this opportunity to express my sincere gratitude to all those who have contributed to the successful completion of this project. Their guidance, support, and encouragement have been invaluable throughout this journey.

I am deeply grateful to the Management and Faculty of Parul University, Faculty of IT and Computer Science, for providing an excellent academic environment, infrastructure, and resources that made this project possible. The department's emphasis on practical, industry-oriented learning has been the foundation of this work.

I would like to express my heartfelt gratitude to **Prof. Vijya Tulsani**, my Internal Project Guide, for her continuous guidance, constructive feedback, and unwavering support throughout the development of this project. Her expertise, patience, and mentorship were instrumental in shaping the direction and quality of this work at every stage.

I extend my sincere thanks to the management and staff of Vermin Industries, Vadodara, for providing access to the inventory data, operational context, and professional environment necessary to make this project practical and industry-relevant. Their cooperation and trust were essential to the successful execution of this project.

I am also thankful to all the faculty members of the Department of MCA for their academic guidance, encouragement, and support throughout the MCA programme. Their teachings and insights have directly contributed to the technical competence demonstrated in this project.

This report covers all aspects of the project lifecycle from system analysis and design through to implementation, testing, and future enhancement planning. It is prepared in accordance with the guidelines prescribed by the Department of MCA, Faculty of IT and Computer Science, Parul University.

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## **About Department of MCA**

### **PARUL University**

Parul University is a legitimate university established under Gujarat Private University Act 2009, after legislation passed by the Government of Gujarat on 26<sup>th</sup> March 2015 giving University status to Parul Group of Institutes functioning under the aegis of Parul Arogya Seva Mandal Trust.

### **Faculty of IT & Computer Science**

Faculty of IT and Computer Science, Parul University has materialized as one of the prime IT education providers at global level. Various departments under Faculty of IT and Computer Science strive in preparing IT-industry ready professionals by means of various skill development courses, vocational courses, co-curricular & extra-curricular activities, industry visits and expert lectures.

### **MCA Department**

The Department of Master of Computer Application at Parul University emphasizes on building professionals in the domain of computer applications by providing necessary environment by means of facilitating suitable blend of technical and non-technical learning experience. The department cultivates students in various curricular, co-curricular and extra-curricular activities in order to produce future system analysts, system designers, system programmers, application programmers, testing professionals, system managers, project managers, researchers and other leading positions in systems/IT department.

The department offers various subjects from diversified technical/non-technical areas such as – core IT domain, management, communication skills, mathematics & logic building and rich pool of elective subjects.

The department of MCA focuses on project-based learning, and hence students are motivated to work on tiny hands-on projects in practical oriented subjects to get better exposure. Moreover, throughout their MCA studies, students are required to work on around 3 mini/major projects in individual/team to get enough confidence on software-development and thereby become industry-read.



## 2. Company Profile

### Short Company Profile

Vermin Industries Private Limited is an emerging industrial automation and electronics solutions company based in Vadodara, Gujarat, India. The company specializes in providing customized automation systems, PLC/SCADA panels, VFDs, servo drives, and motion control solutions for industries such as pharmaceuticals and technical fiber manufacturing. With a strong focus on innovation, reliability, and efficiency, Vermin Industries helps businesses improve productivity, safety, and operational performance through advanced automation technologies.

### Type of Company

Vermin Industries Pvt. Ltd. is an automation and electronics engineering company that designs and provides industrial automation systems used in manufacturing industries like pharmaceuticals and technical fiber plants. The company develops customized automation solutions that help industries improve productivity, safety, and process efficiency

### Key Features:

- Strong focus on quality control
- Skilled workforce and technical expertise
- Customer-centric approach
- Use of modern machinery and tools

### Strengths:

- Strategic location in Vadodara (industrial hub)
- Efficient supply chain management
- Commitment to innovation and quality
- Experienced management team

### 3. Project Profile

#### 3.1 Project Definition

The project is to design and develop a Vermin Industries Inventory Analytics System — a full- stack web-based application for managing inventory data, procurement performance, supplier relationships, and stock health. The system includes an interactive 7-page Power BI dashboard, a MySQL database backend, and a Streamlit web application for real-time data management and SQL-based querying.

#### 3.2 Project Description

The Vermin Industries Inventory Analytics System is a comprehensive data analytics platform that digitises and streamlines inventory management. The system manages data across four core relational tables covering suppliers, components, purchase orders, and stock movements. It provides real-time visibility into procurement performance, supplier quality benchmarking across India and Korea, and stock health monitoring.

The system can be understood through a real-life workflow scenario. When a procurement officer places a purchase order, the data is immediately stored in the centralized database. The inventory stock is automatically updated once the delivery is received. The system then calculates key performance indicators such as delivery rate, stock availability, and supplier performance. This eliminates manual tracking and reduces human errors.

Additionally, the system provides decision-making support by identifying critical stock items and generating alerts. This ensures that the organization can maintain optimal inventory levels and avoid production delays.

#### 3.3 Existing System / Work Environment

Prior to the development of this system, inventory management at Vermin Industries relied on the following manual and semi-digital methods:

- Manual Excel spreadsheets with no relational structure or data integrity enforcement
- No centralized database — data scattered across multiple individual Excel files
- No automated stock alerts or reorder level notifications for critical components
- No structured supplier performance benchmarking between India and Korea vendors
- No interactive dashboard for management-level KPI reporting and decision-making
- No ability to run analytical queries across multiple data tables simultaneously

### 3.4 Problem Statements

The existing system presented the following key challenges:

- **Data Integrity:** Multiple unlinked Excel files caused data duplication, inconsistencies, and version conflicts across departments.
- **Inefficient Stock Tracking:** No automated mechanism existed to classify stock status or flag components falling below reorder levels.
- **No Centralised Platform:** Managers had no single interface to view procurement, supplier, and stock KPIs together.
- **Lack of Real-Time Insights:** No dashboards or calculated metrics were available for management reporting.
- **No Supplier Benchmarking:** Performance comparison between India and Korea suppliers was done manually and infrequently.
- **Limited Accessibility:** Inventory data was only accessible via physical file access with no web-based interface.

### 3.5 Need for New System

The limitations of the existing system created a clear and urgent need for a new, centralised inventory analytics platform. The new system was required to:

- Establish a normalised relational database to eliminate data duplication and ensure integrity
- Automate stock status classification and generate alerts for critical inventory items
- Provide an interactive web-based interface accessible from any authorised device
- Deliver a multi-page Power BI dashboard with real-time KPIs for management reporting
- Enable live SQL query execution for ad-hoc analytical reporting
- Allow CRUD operations on all inventory data tables through a controlled interface

### 3.6 Proposed System and Features

The proposed Inventory Analytics System addresses all identified problems through the following features

#### Core Analytics Features

- MySQL relational database with 4 normalised tables and 3 enforced foreign key relationships
- 7-page interactive Power BI dashboard with 22 custom DAX measures and cross-page slicers
- Streamlit web application with multi-tab interface for all inventory management tasks
- Full CRUD operations — Add, Edit, Delete records across all four data tables
- Live SQL query runner using in-memory SQLite engine with CSV export capability

- Automated stock status classification into Critical, Low, Normal, and Excess categories
- India vs Korea supplier comparison with ratings, lead time, and order volume benchmarking
- Overall Inventory Health Score calculated using a weighted composite DAX formula

### 3.7 Scope

The scope of this project covers the complete data lifecycle from data ingestion through to interactive analytics and data management. The system is designed for internal use by Vermin Industries inventory management personnel. Data coverage spans June to December 2024, encompassing 120 purchase orders across 17 components supplied by 12 vendors from India (6) and Korea (6). Future scope includes role-based access control and multi-warehouse support.

### 3.8 Outcomes

- 94.17% delivery rate identified — 113 out of 120 purchase orders delivered successfully
- 42 critical stock records flagged automatically for immediate procurement action
- Korea suppliers rated higher at 4.28 vs India at 4.11 with 2 days faster delivery
- Overall Inventory Health Score of 84% established as measurable performance baseline
- Full-stack application delivered with 707+ lines of production-quality Python code
- Centralised platform replacing fragmented Excel-based manual tracking.

### 3.9 Tools & Technology Used

The Inventory Analytics project is developed using the following technologies:

#### Core Development

- Python 3.10+ - primary language
- Streamlit - web application framework

#### Database

- MySQL 8.0 — relational database (4 tables, 3 FK relationships)
- SQLite (in-memory) — live SQL query runner inside Streamlit

### Business Intelligence

- Microsoft Power BI Desktop — 7-page dashboard, DAX measures, cross-page slicers.

### Data Processing & Analysis

- Pandas — data cleaning, CRUD operations, DataFrame management
- NumPy — numerical processing

### Data Storage & Format

- Microsoft Excel (.xlsx) — source data file and export format

### Development Environment

- VS Code / PyCharm — IDE

### Operating System / Platform

- Windows 10/11 (64-bit) — host OS
- Web Browser (Chrome / Edge) — application access

## 3.10 Project Plan

| Phase | Activity                                    | Duration | Status    |
|-------|---------------------------------------------|----------|-----------|
| 1     | Requirement gathering & data collection     | Week 1   | Completed |
| 2     | MySQL database design & normalization       | Week 2   | Completed |
| 3     | Power BI dashboard development (7 pages)    | Week 3–4 | Completed |
| 4     | Streamlit app — core data management module | Week 5   | Completed |
| 6     | Documentation & final report                | Week 7–8 | Completed |

## 4. Requirement Analysis

**Requirement Analysis** is a crucial phase in the Software Development Life Cycle (SDLC). It involves understanding the needs of users, feasibility, identifying functional modules, and determining the hardware/software requirements for successful implementation.

### 4.1 Feasibility Study

The project is technically feasible. Python, Streamlit, MySQL, and Power BI are all well-documented, widely supported tools with active communities. The hardware requirements are minimal and available on standard development machines.

#### Operational Feasibility

The system is operationally feasible. The Streamlit interface is browser-based and requires no installation by end users beyond a web browser. Inventory managers with basic computer literacy can operate the system with minimal training.

#### Economic Feasibility

The project is economically feasible. All tools used — Python, Streamlit, MySQL Community Edition, and Power BI Desktop — are either free or available under free-tier licenses. The system replaces a costly and error-prone manual process, providing strong return on investment through reduced data errors and faster reporting.

#### Schedule Feasibility

The project was completed within an 8-week timeline as planned. The modular development approach — database first, then dashboard, then web application — allowed systematic testing at each stage.

### 4.2 Users of the System

| User Type           | Description                                      | Access Level |
|---------------------|--------------------------------------------------|--------------|
| Inventory Manager   | Primary user — manages stock, POs, and suppliers | Full access  |
| Procurement Officer | Raises and tracks purchase orders                | Full access  |
| Senior Management   | Views KPI dashboards and executive summary       | Full access  |

|              |                                             |                    |
|--------------|---------------------------------------------|--------------------|
| System Admin | Manages data files and system configuration | Full system access |
|--------------|---------------------------------------------|--------------------|

### 4.3 Modules

The system is divided into the following functional modules:

#### Module 1 — Data Management

Provides tabbed views for all four data tables: Suppliers, Components\_Master, Purchase\_Orders, and Inventory\_Stock. Supports full CRUD operations with form validation and auto-generated IDs.

#### Module 2 — SQL Query Runner

Allows users to write and execute custom SQL queries against all data tables using an in- memory SQLite engine. Results are displayed in a data grid and can be exported as CSV.

#### Module 3 — Power BI Dashboard

Embeds the published Power BI report via an iframe URL. Provides the 7-page interactive dashboard covering Inventory Overview, Purchase Orders, Supplier Intelligence, Stock and Consumption, Reorder Alerts, India vs Korea comparison, and Executive Summary.

#### Module 4 — Data Processing

Handles data loading from Excel, cleaning operations (deduplication, null handling, whitespace stripping), and saving updates back to the Excel file for Power BI refresh.

### 4.4 Process Model

The project follows the Iterative Development Model. Each module was developed, tested, and integrated incrementally. The development progressed through the following iterations:

- Iteration 1: Database design and data preparation (MySQL + Excel)
- Iteration 2: Power BI dashboard with DAX measures and slicers
- Iteration 3: Core Streamlit application with data management tabs
- Iteration 4: SQL query runner and CSV export functionality
- Iteration 5: Integration testing and final documentation

### 4.5 Hardware and Software Requirements

The Inventory Analytics System was developed and tested on a standard development machine. The following section outlines the minimum and recommended hardware requirements, as well as all software dependencies required to develop, run, and use the system.

Table 1 — Hardware Requirements

| Component | Minimum                        | Recommended                 |
|-----------|--------------------------------|-----------------------------|
| Processor | Intel Core i3 / AMD equivalent | Intel Core i5 or above      |
| RAM       | 4 GB                           | 8 GB or above               |
| Storage   | 20 GB HDD                      | 50 GB SSD                   |
| Display   | 1280 × 720 resolution          | 1920 × 1080 (Full HD)       |
| Network   | Not required (local)           | Required for Power BI embed |
| OS        | Windows 10 (64-bit)            | Windows 11 (64-bit)         |

Table 2 — Software Requirements

| Software                   | Version           | Purpose                            |
|----------------------------|-------------------|------------------------------------|
| Python                     | 3.10 or above     | Primary programming language       |
| Streamlit                  | Latest stable     | Web application framework          |
| MySQL Community Edition    | 8.0               | Relational database server         |
| Microsoft Power BI Desktop | Latest version    | 7-page interactive dashboard       |
| Microsoft Excel            | 2016 / Office 365 | Source data file and export format |
| Pandas                     | Latest stable     | Data cleaning and CRUD operations  |
| NumPy                      | Latest stable     | Numerical processing               |
| Matplotlib / Seaborn       | Latest stable     | EDA charts and visualisations      |
| SQLite (in-memory)         | Built into Python | Live SQL query runner              |



|                             |                |                        |
|-----------------------------|----------------|------------------------|
| <b>VS Code / PyCharm</b>    | Any version    | Development IDE        |
| <b>Google Chrome / Edge</b> | Latest version | Browser for app access |

#### 4.6 Use Cases

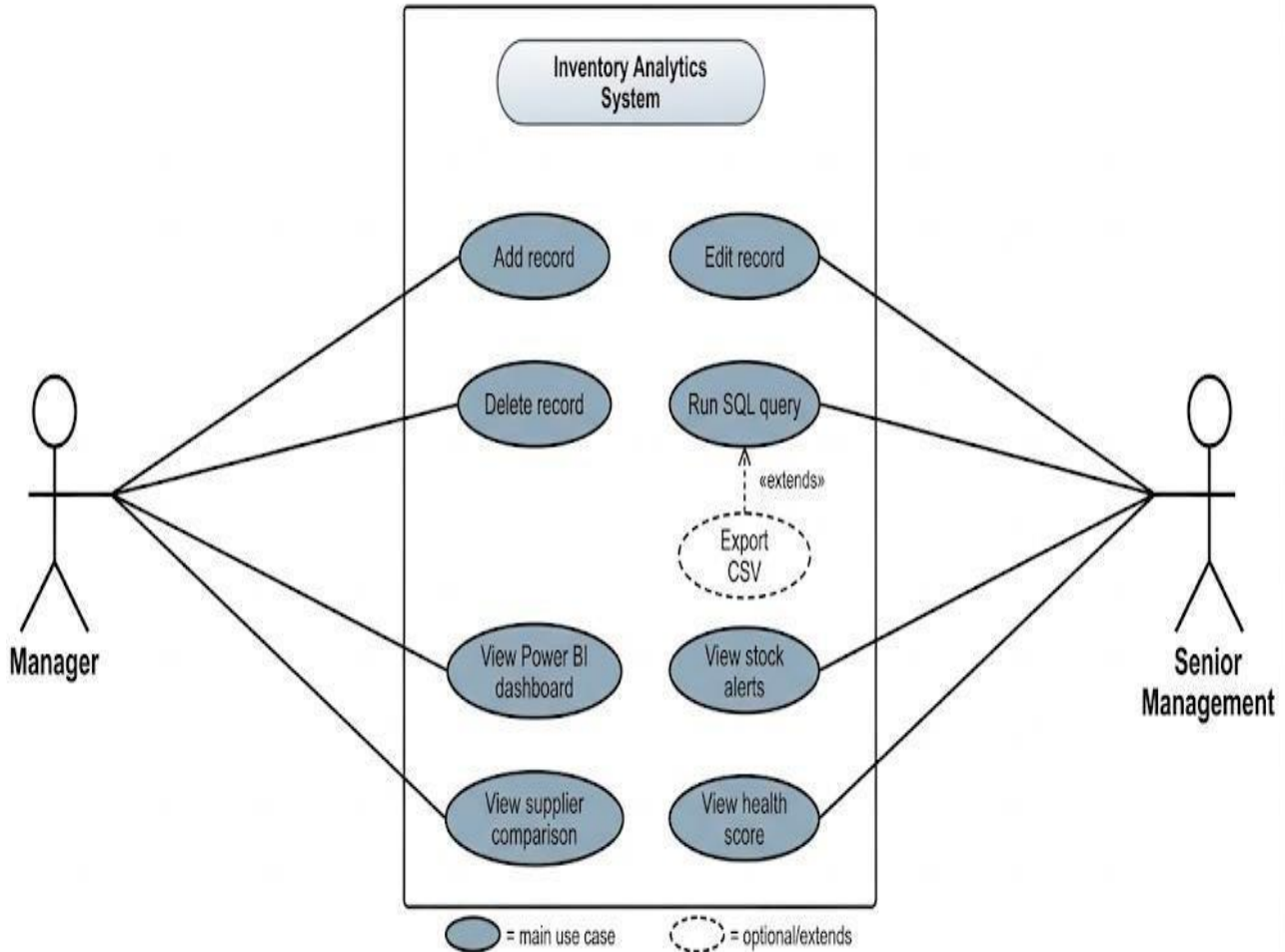
| UC No. | Use Case Name       | Actor                 | Description                                | Outcome           |
|--------|---------------------|-----------------------|--------------------------------------------|-------------------|
| UC-01  | View Inventory Data | Manager               | View all 4 data tables in tabbed interface | Data displayed    |
| UC-02  | Add Record          | Manager               | Insert new row into any data table         | Record added      |
| UC-03  | Edit Record         | Manager               | Modify existing row in any data table      | Record updated    |
| UC-04  | Delete Record       | Manager               | Remove a row from any data table           | Record deleted    |
| UC-05  | Run SQL Query       | Manager               | Execute custom SQL and view results        | Results displayed |
| UC-06  | Export Results      | Manager               | Download SQL query results as CSV          | File downloaded   |
| UC-07  | View Dashboard      | Manager / Senior Mgmt | View 7-page Power BI dashboard             | KPIs displayed    |
| UC-08  | Filter Dashboard    | Manager               | Apply slicers for country/category/status  | Filtered view     |
| UC-09  | Download Excel      | Manager               | Download live data Excel file              | File downloaded   |

#### 4.7 Use Case Diagram

The Use Case Diagram represents the interactions between the system actors and the core use cases of the Inventory Analytics System.

**Primary Actors:**

- **Manager** — can access all system features including data management, SQL queries, and dashboard



- **Senior Management** — can view the Power BI dashboard and KPI reports

The **Inventory Analytics System** use case diagram shows two actors :- **Manager** and **Senior Management** - interacting with the system. The Manager has full access to all 9 use cases including data operations (Add, Edit, Delete records, Run SQL query) and analytical views, while Senior Management can only access the 4 reporting/view features. **Export CSV** is an optional extension of Run SQL query, shown with a dashed ellipse and «extends» relationship.

## 5. Design

### 5.1 Use Case Scenarios

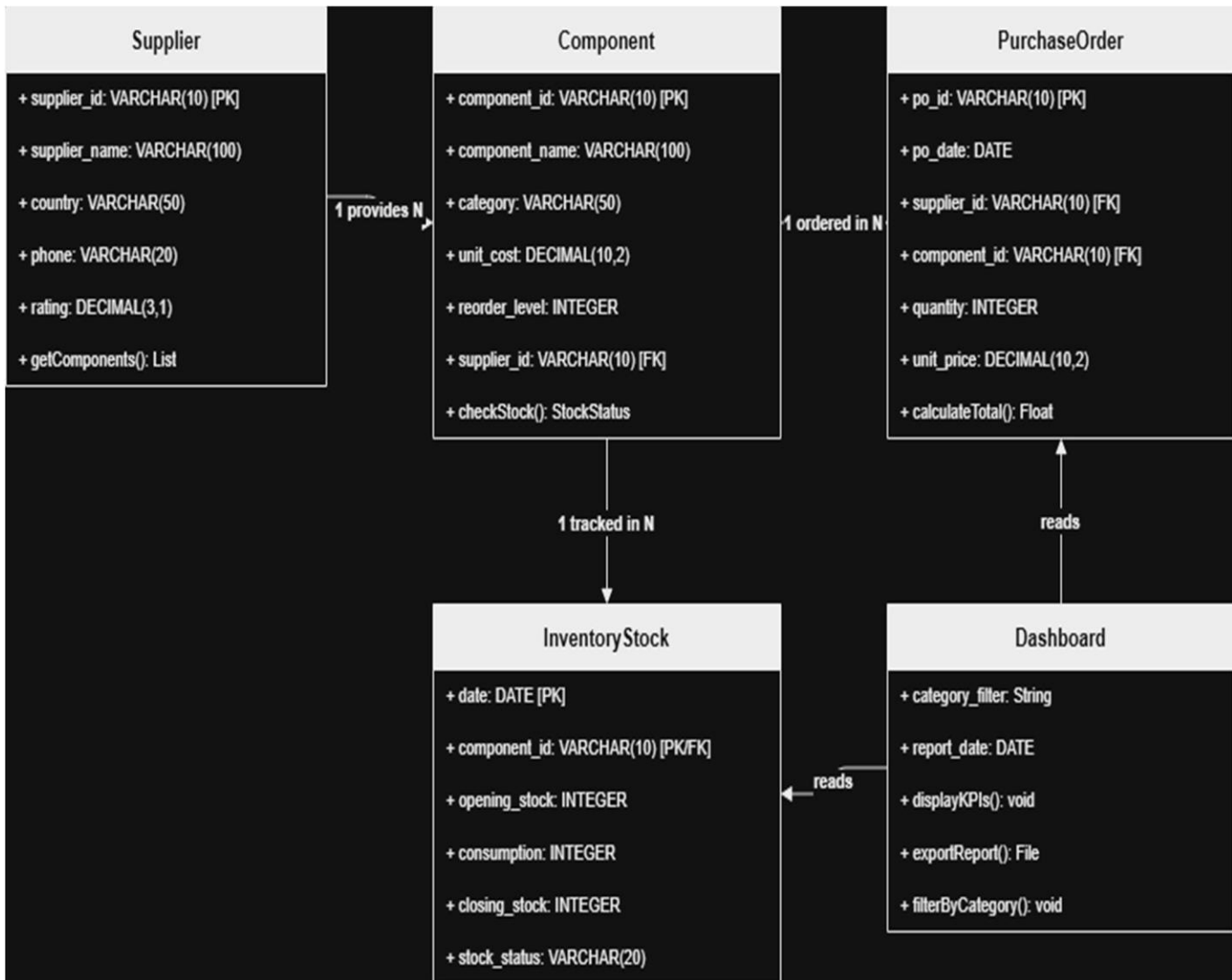
| <b>Use Case ID</b>    | UC-01                                                                                                                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name</b>  | View Inventory Data                                                                                                                                                                                          |
| <b>Actor</b>          | Manager                                                                                                                                                                                                      |
| <b>Precondition</b>   | Application is running and data is loaded                                                                                                                                                                    |
| <b>Main Flow</b>      | <ol style="list-style-type: none"> <li>1. User opens the application</li> <li>2. System loads all 4 data tables</li> <li>3. User selects a tab</li> <li>4. System displays records in a data grid</li> </ol> |
| <b>Alternate Flow</b> | If data file is missing — system displays an error message                                                                                                                                                   |
| <b>Postcondition</b>  | Records are displayed in the selected tab                                                                                                                                                                    |

| <b>Use Case ID</b>    | UC-05                                                                                                                                                                                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name</b>  | Run SQL Query                                                                                                                                                                                                                                                                                            |
| <b>Actor</b>          | Manager                                                                                                                                                                                                                                                                                                  |
| <b>Precondition</b>   | Data tables are loaded into session                                                                                                                                                                                                                                                                      |
| <b>Main Flow</b>      | <ol style="list-style-type: none"> <li>1. User navigates to SQL Query tab</li> <li>2. User types SQL statement</li> <li>3. User clicks Run Query</li> <li>4. System loads tables into SQLite</li> <li>5. System executes query and displays results</li> <li>6. User optionally downloads CSV</li> </ol> |
| <b>Alternate Flow</b> | If SQL is invalid — system displays error message from SQLite engine                                                                                                                                                                                                                                     |
| <b>Postcondition</b>  | Query results displayed; optional CSV file available for download                                                                                                                                                                                                                                        |

**View & Monitor Inventory Data** A warehouse manager opens the dashboard and navigates through the Suppliers, Components, Purchase Orders, and Inventory Stock tabs to get a real-time overview of all inventory records and supplier information.

## 5.2 Diagrams

### 5.2.1 UML / DFD



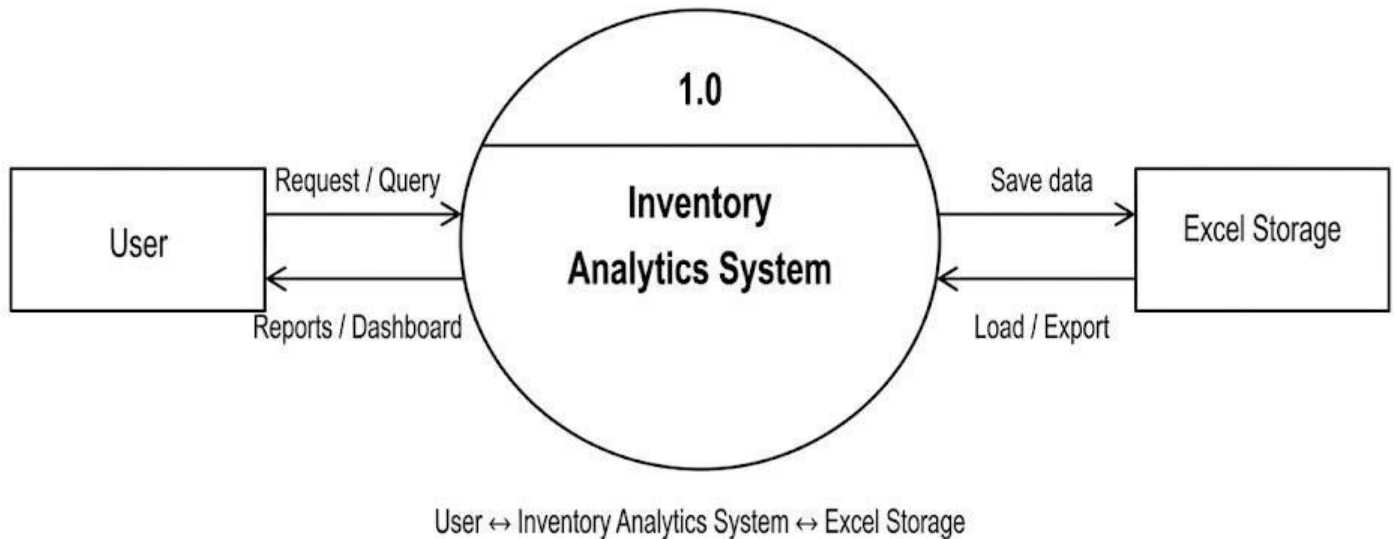
This is a **UML Class Diagram** for the **Inventory Analytics System**, showing 5 classes and their relationships.

**Supplier** supplies components to **Component** (1 provides N relationship), and **Component** is tracked in **InventoryStock** (1 tracked in N). **Component** is also ordered through **PurchaseOrder** (1 ordered in N). The **Dashboard** class reads data from both **InventoryStock** and **PurchaseOrder** to display KPIs and generate reports.

Each class contains attributes (fields like IDs, names, dates) and methods (like `checkStock()`, `calculateTotal()`, `displayKPIs()`), following standard object-oriented design for managing inventory, suppliers, stock levels, and purchase orders.

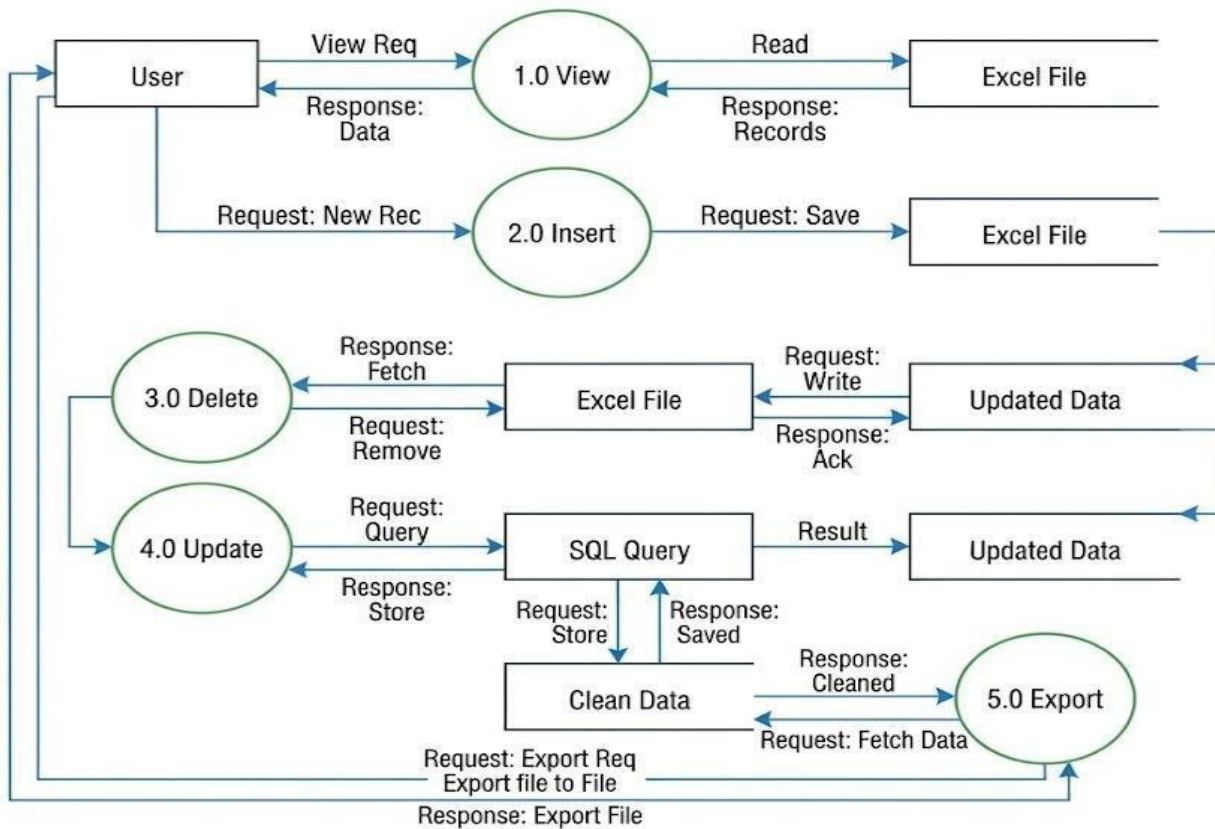
### 5.2.1 DFD Diagrams

**Level 0 DFD — Context Diagram**



This is a Level 0 DFD (Context Diagram) for the Inventory Analytics System, showing the system's interaction with two external entities.

The User sends requests/queries to the system and receives back reports and dashboard views. The Excel Storage acts as the data store — the system saves data to it and loads/exports data from it. The central Inventory Analytics System process sits in the middle, handling all data flow between the user and the storage. It gives a high-level overview of the entire system in a single diagram with no internal details shown.

**Level 1 DFD****First Level DFD**

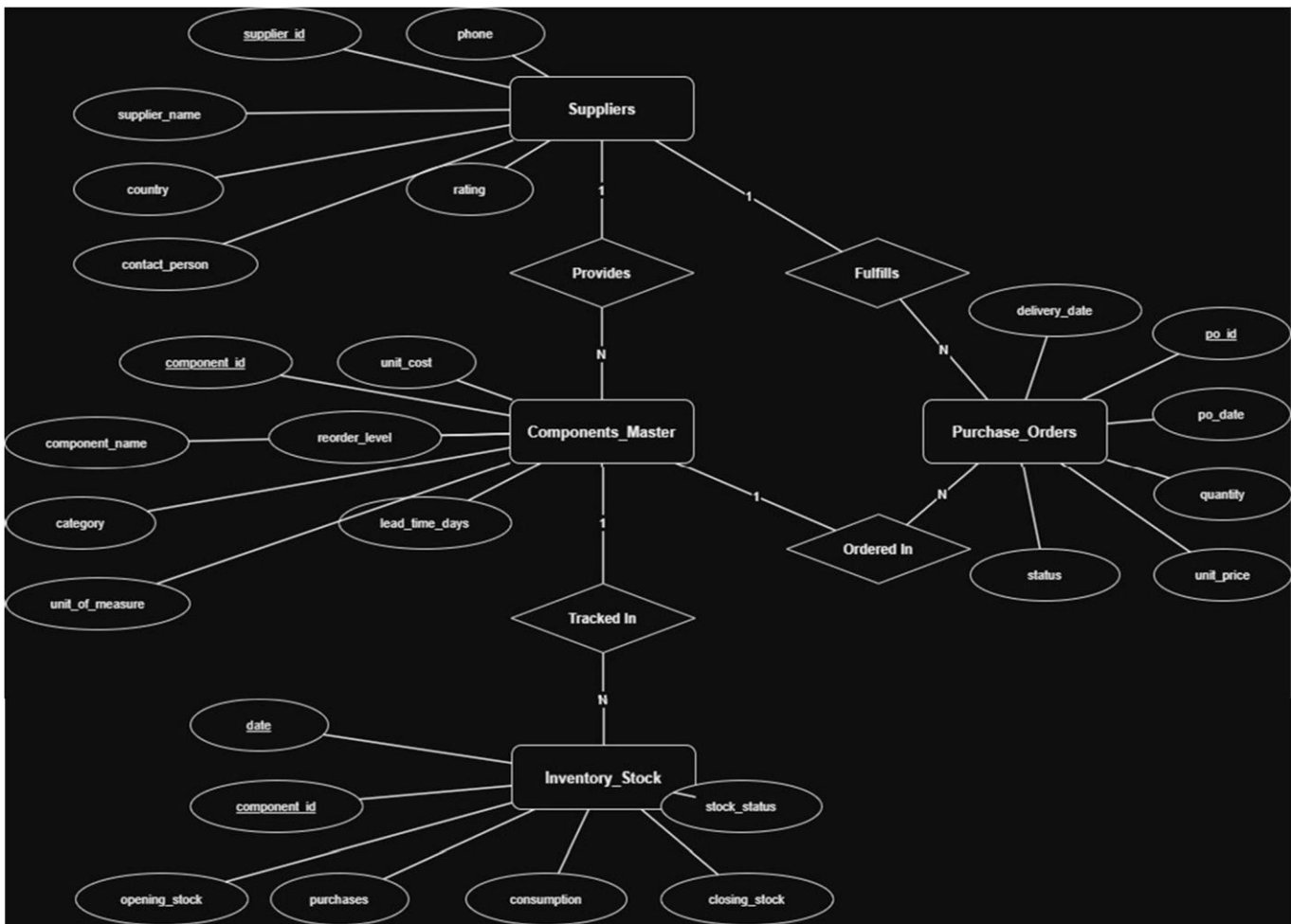
This is a Level 1 DFD (Data Flow Diagram) for the Inventory System, breaking down the internal processes in detail across 5 sub-processes.

1.0 View Data — the User sends input, the process reads from the Excel File and displays records back to the User. 2.0 Insert Record — the User sends a new record which gets saved into the Excel File. 3.0 Delete Record — fetches a record from the Excel File and removes it. 4.0 Update Record — sends a query to the SQL Query store, retrieves raw data, processes it, and writes the result to Updated Data. Finally, 5.0 Export Data — takes cleaned data from the Clean Data store and exports the final file back to the User.

## 5.2.2 Entity Relationship Diagram

An **Entity-Relationship (ER) diagram** is a graphical tool used to model the structure of a database. It identifies the main entities in a system, such as people, objects, or concepts, and represents them as rectangles. Each entity has attributes, which describe its properties and are shown as ovals connected to the entity. Relationships, represented by diamonds or lines, illustrate how entities are linked to each other — for example, a “student” *enrolls in* a “course.”

ER diagrams are essential during the database design phase because they provide a clear, visual understanding data of how flows and connects within a system.



This is an Entity-Relationship (ER) Diagram for the Inventory Analytics System, showing 4 entities and their relationships.

Suppliers has attributes like `supplier_id`, `supplier_name`, `country`, `phone`, `rating`, and `contact_person`. It connects to `Components_Master` and to `Purchase_Orders`.

`Components_Master` holds attributes like `component_id`, `component_name`, `category`, `unit_cost`, `reorder_level`, `lead_time_days`, and `unit_of_measure`.

`Inventory_Stock` tracks `date`, `component_id`, `opening_stock`, `purchases`, `consumption`, `closing_stock`, and `stock_status` — recording daily stock movement per component.

`Purchase_Orders` stores `po_id`, `po_date`, `delivery_date`, `quantity`, `unit_price`, and `status`.



### 5.3 Data Dictionary

**Table 1: SUPPLIERS**

| Column Name    | Data Type    | Constraint  | Allow Null | Description                            |
|----------------|--------------|-------------|------------|----------------------------------------|
| supplier_id    | VARCHAR(10)  | PRIMARY KEY | NO         | Unique supplier identifier e.g. S0001  |
| supplier_name  | VARCHAR(100) | NOT NULL    | NO         | Full name of the supplier company      |
| country        | VARCHAR(50)  | NOT NULL    | NO         | Country of origin — India or Korea     |
| contact_person | VARCHAR(100) | —           | YES        | Name of the primary contact person     |
| phone          | VARCHAR(20)  | —           | YES        | Contact phone number of the supplier   |
| rating         | DECIMAL(3,2) | NOT NULL    | NO         | Supplier performance rating out of 5.0 |

**Table 2: COMPONENTS\_MASTER**

| Column Name    | Data Type     | Constraint  | Allow Null | Description                                              |
|----------------|---------------|-------------|------------|----------------------------------------------------------|
| component_id   | VARCHAR(10)   | PRIMARY KEY | NO         | Unique component identifier e.g. COMP001                 |
| component_name | VARCHAR(100)  | NOT NULL    | NO         | Descriptive name of the component                        |
| category       | VARCHAR(50)   | NOT NULL    | NO         | Category: Screws, Locks, Connectors, Wires, Push Buttons |
| supplier_id    | VARCHAR(10)   | FOREIGN KEY | NO         | References SUPPLIERS(supplier_id)                        |
| unit_cost      | DECIMAL(10,2) | NOT NULL    | NO         | Cost per unit in INR                                     |
| reorder_level  | INT           | NOT NULL    | NO         | Minimum stock quantity before reorder is triggered       |
| lead_time_days | INT           | —           | YES        | Number of days to receive stock after placing order      |

**Table 3: PURCHASE\_ORDERS**

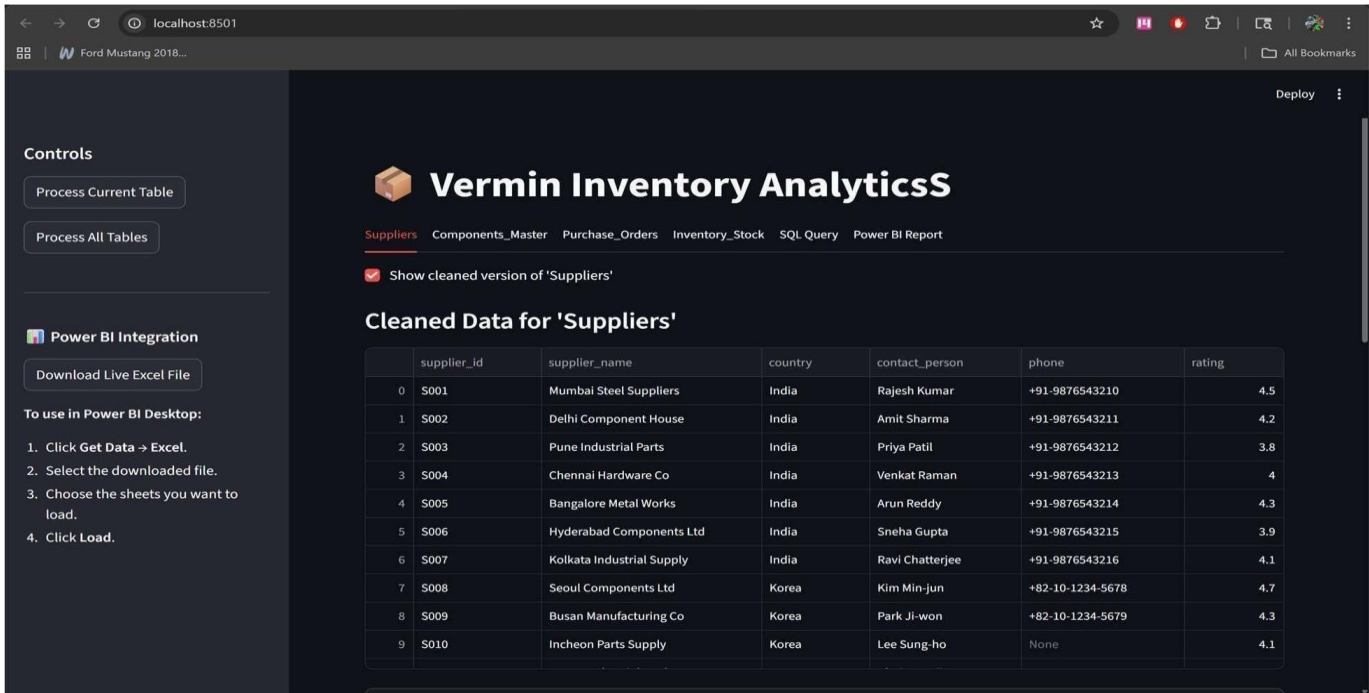
| Column Name   | Data Type     | Constraint  | Allow Null | Description                                     |
|---------------|---------------|-------------|------------|-------------------------------------------------|
| po_id         | VARCHAR(10)   | PRIMARY KEY | NO         | Unique purchase order ID e.g. PO0001            |
| component_id  | VARCHAR(10)   | FOREIGN KEY | NO         | References COMPONENTS_MASTER(component_id)      |
| supplier_id   | VARCHAR(10)   | NOT NULL    | NO         | Supplier reference (denormalised for reporting) |
| quantity      | INT           | NOT NULL    | NO         | Number of units ordered                         |
| unit_price    | DECIMAL(10,2) | NOT NULL    | NO         | Price per unit at time of order in INR          |
| po_date       | DATE          | NOT NULL    | NO         | Date when the purchase order was placed         |
| delivery_date | DATE          | —           | YES        | Actual date the order was delivered             |
| status        | VARCHAR(20)   | NOT NULL    | NO         | Order status — Delivered or Pending             |

**Table 4: INVENTORY\_STOCK**

| Column Name   | Data Type   | Constraint  | Allow Null | Description                                 |
|---------------|-------------|-------------|------------|---------------------------------------------|
| component_id  | VARCHAR(10) | FOREIGN KEY | NO         | References COMPONENTS_MASTER(component_id)  |
| stock_date    | DATE        | NOT NULL    | NO         | Date of this stock movement record          |
| opening_stock | INT         | NOT NULL    | NO         | Stock quantity at start of the period       |
| purchases     | INT         | —           | YES        | Units received from supplier in this period |
| consumption   | INT         | —           | YES        | Units consumed or used in this period       |
| closing_stock | INT         | NOT NULL    | NO         | Stock quantity at end of the period         |
| stock_status  | VARCHAR(20) | NOT NULL    | NO         | Status: Critical / Low / Normal / Excess    |

The data dictionary defines the structure of 4 tables — **Suppliers**, **Components\_Master**, **Purchase\_Orders**, and **Inventory\_Stock** used in the Inventory Analytics System. Each table uses a primary key for unique identification, and foreign keys link related tables together, such as `supplier_id` and `component_id` connecting across tables.

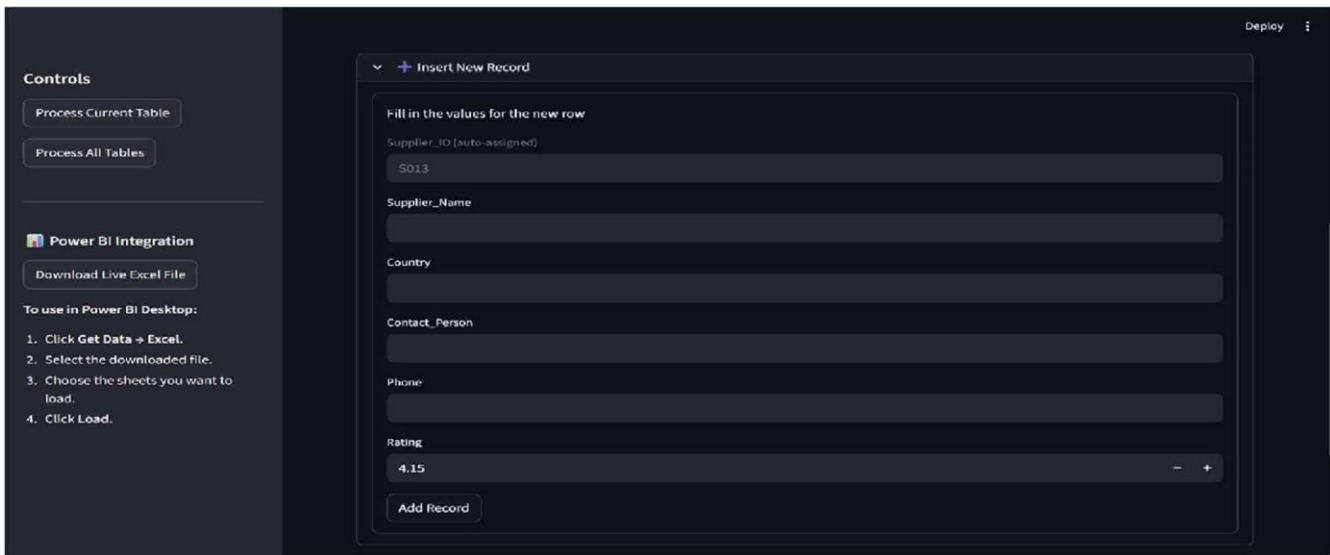
## 6. Implementation



The screenshot shows the Vermin Inventory Analytics dashboard. On the left, there are controls for 'Process Current Table' and 'Process All Tables', and a 'Power BI Integration' section with a 'Download Live Excel File' button. The main area displays the 'Suppliers' table with a toggle for 'Show cleaned version of Suppliers'. The table shows 10 rows of supplier data.

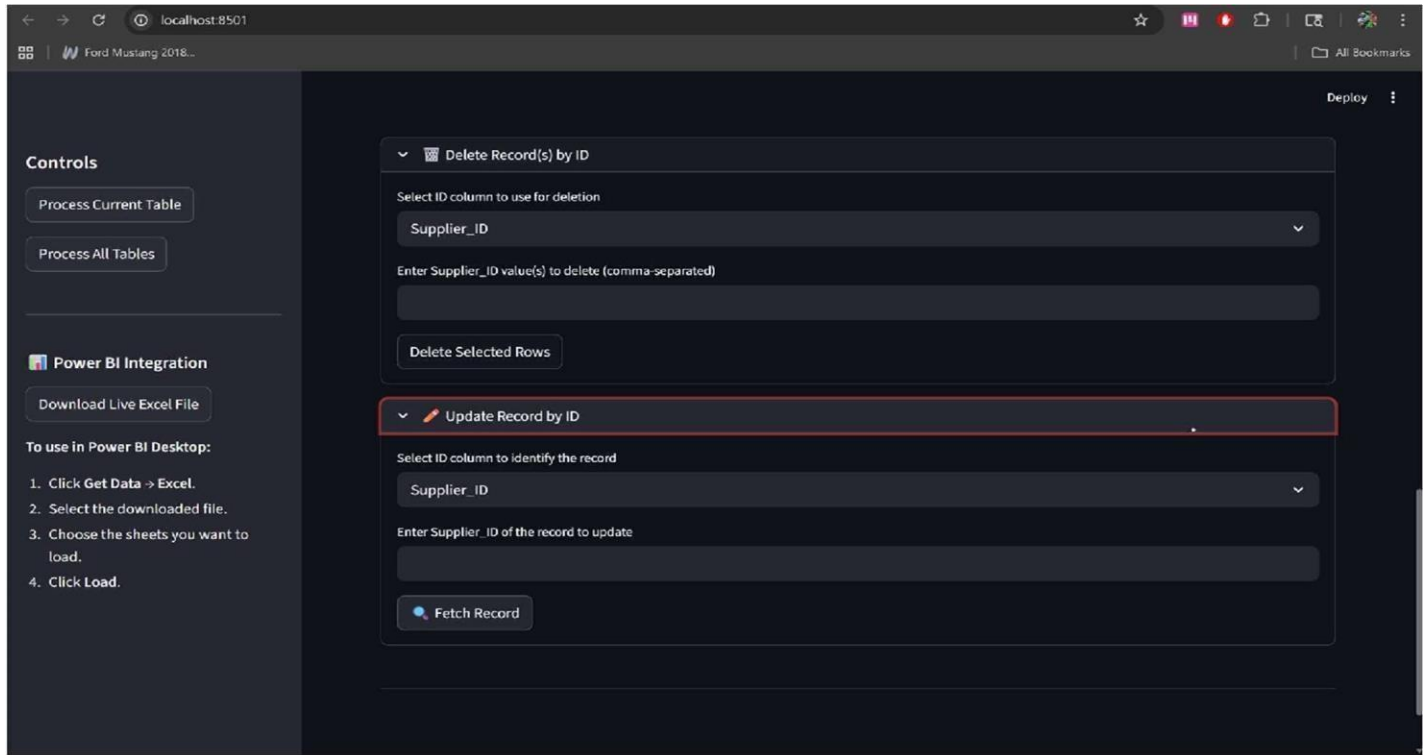
|   | supplier_id | supplier_name             | country | contact_person  | phone            | rating |
|---|-------------|---------------------------|---------|-----------------|------------------|--------|
| 0 | S001        | Mumbai Steel Suppliers    | India   | Rajesh Kumar    | +91-9876543210   | 4.5    |
| 1 | S002        | Delhi Component House     | India   | Amit Sharma     | +91-9876543211   | 4.2    |
| 2 | S003        | Pune Industrial Parts     | India   | Priya Patil     | +91-9876543212   | 3.8    |
| 3 | S004        | Chennai Hardware Co       | India   | Venkat Raman    | +91-9876543213   | 4      |
| 4 | S005        | Bangalore Metal Works     | India   | Arun Reddy      | +91-9876543214   | 4.3    |
| 5 | S006        | Hyderabad Components Ltd  | India   | Sneha Gupta     | +91-9876543215   | 3.9    |
| 6 | S007        | Kolkata Industrial Supply | India   | Ravi Chatterjee | +91-9876543216   | 4.1    |
| 7 | S008        | Seoul Components Ltd      | Korea   | Kim Min-jun     | +82-10-1234-5678 | 4.7    |
| 8 | S009        | Busan Manufacturing Co    | Korea   | Park Ji-won     | +82-10-1234-5679 | 4.3    |
| 9 | S010        | Incheon Parts Supply      | Korea   | Lee Sung-ho     | None             | 4.1    |

Vermin Inventory Analytics displays a clean, dark-themed dashboard showing the Suppliers table with details like supplier ID, name, country, contact person, phone, and rating. It includes a "Show cleaned version" toggle that filters and displays sanitized data for better accuracy.



The screenshot shows the 'Insert New Record' form. It has a title 'Insert New Record' and a subtitle 'Fill in the values for the new row'. The form contains input fields for 'Supplier\_ID (auto-assigned)' (with value S013), 'Supplier\_Name', 'Country', 'Contact\_Person', 'Phone', and 'Rating' (with value 4.15). There is an 'Add Record' button at the bottom.

The second screen shows an "Insert New Record" form where users can add a new supplier by filling in fields like Supplier Name, Country, Contact Person, Phone, and Rating, with the Supplier ID being auto-assigned by the system.



This screen shows two CRUD operation panels — "Delete Record(s) by ID" and "Update Record by ID." Users can select the Supplier\_ID column, enter comma-separated IDs to delete rows, or fetch a specific record by ID to update it directly from the dashboard.

The screenshot shows the Vermin Inventory AnalyticsS web application. The left sidebar contains controls for processing tables and Power BI integration. The main area is titled "Run SQL Queries on Your Data" and lists available tables: Suppliers, Components\_Master, Purchase\_Orders, and Inventory\_Stock. A text input field contains the SQL query "Select \* from supplie;". Below the input field is a "Run Query" button. At the bottom, a red error message states: "SQL Error: Execution failed on sql 'Select \* from supplie; ': no such table: supplie".

The SQL Query tab allows users to run custom SQL queries directly on their inventory data across four available tables :- Suppliers, Components\_Master, Purchase\_Orders, and Inventory\_Stock. It also displays real-time error feedback, as seen with the SQL error message at the bottom for an incorrectly typed table name.

The screenshot shows the Vermin Inventory AnalyticsS web application with the SQL Query tab. The left sidebar is the same as in the previous screenshot. The main area shows the same list of available tables. The text input field now contains the correct SQL query: "Select \* from suppliers;". The "Run Query" button is highlighted with a red border. Below the input field, the "Query Result" section displays a table with 7 columns: Supplier\_ID, Supplier\_Name, Country, Contact\_Person, Phone, and Rating. The table contains 5 rows of data.

|   | Supplier_ID | Supplier_Name          | Country | Contact_Person | Phone          | Rating |
|---|-------------|------------------------|---------|----------------|----------------|--------|
| 0 | S001        | Mumbai Steel Suppliers | India   | Rajesh Kumar   | +91-9876543210 | 4.5    |
| 1 | S002        | Delhi Component House  | India   | Amit Sharma    | +91-9876543211 | 4.2    |
| 2 | S003        | Pune Industrial Parts  | India   | Priya Patil    | +91-9876543212 | 3.8    |
| 3 | S004        | Chennai Hardware Co    | India   | Venkat Raman   | +91-9876543213 | 4      |
| 4 | S005        | Bangalore Metal Works  | India   | Arun Reddy     | +91-9876543214 | 4.3    |

This screen shows a successful SQL query execution where "Select \* from suppliers;" retrieves all supplier records, displaying the full Query Result table with columns like Supplier\_ID, Supplier\_Name, Country, Contact\_Person, Phone, and Rating in real time.

The screenshot shows the application interface at localhost:8501. On the left, the 'Controls' section has buttons for 'Process Current Table' and 'Process All Tables' (highlighted with a red box). Below it, the 'Power BI Integration' section includes a 'Download Live Excel File' button and instructions for using the application in Power BI Desktop. The main area displays a table of suppliers:

| ID | Table Name        | Table Name        |
|----|-------------------|-------------------|
| 0  | Suppliers         | suppliers         |
| 1  | Components_Master | components_master |
| 2  | Purchase_Orders   | purchase_orders   |
| 3  | Inventory_Stock   | inventory_stock   |

Below the table, there is a text input field for an SQL query containing "Select \* from suppliers;". A 'Run Query' button is present. A green success message states: "All tables processed and packaged." Below this message is a 'Download All Tables as ZIP' button.

This screen shows the "Process All Tables" feature in action, where all four inventory tables are processed and packaged together. A green success message confirms "All tables processed and packaged," with an option to download all tables as a single ZIP file.

The screenshot shows the application interface at localhost:8501. On the left, the 'Controls' section has buttons for 'Process Current Table' and 'Process All Tables'. Below it, the 'Power BI Integration' section includes a 'Download Live Excel File' button and instructions for using the application in Power BI Desktop. The main area displays the 'Vermin Inventory AnalyticsS' logo and a section titled 'Embedded Power BI Report'. It includes a warning message: "Local .pbix files cannot be embedded directly in any web application." and instructions on how to embed a Power BI report:

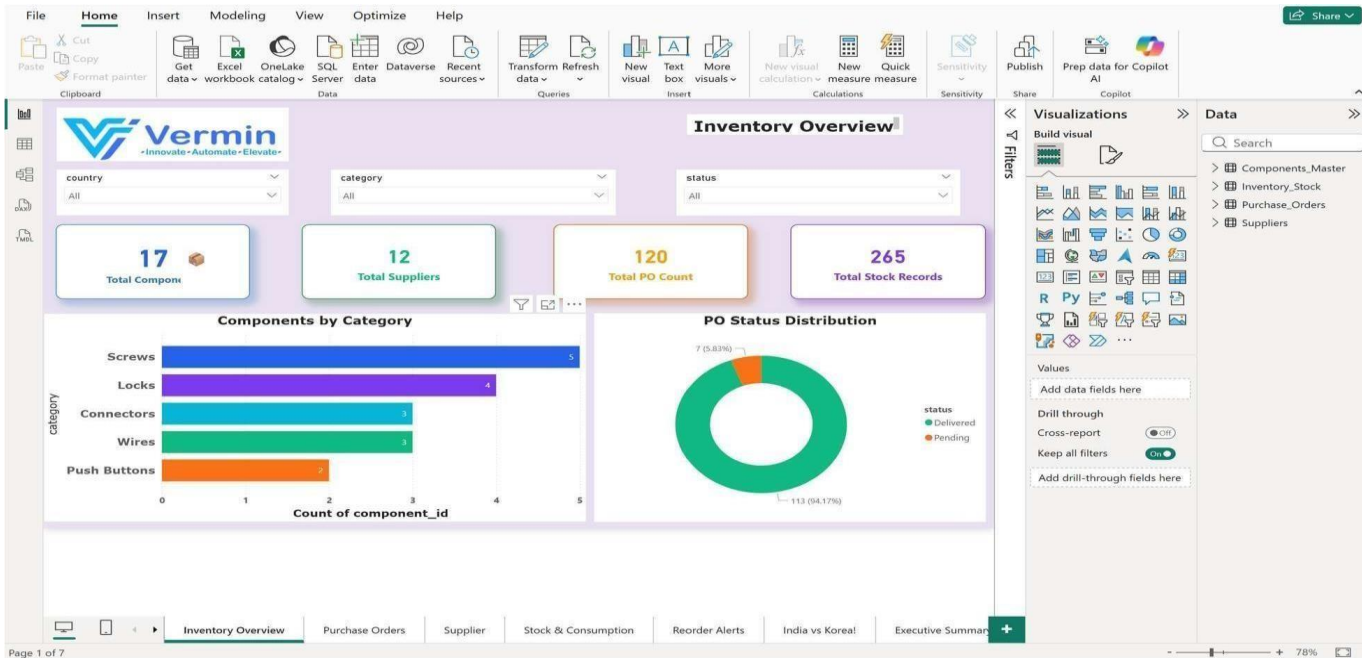
1. Publish your report to the Power BI service (app.powerbi.com)
2. Get a public embed URL from the published report
3. Paste that URL below

Below this, there is a section titled 'Instructions to get your embed URL:' with the following steps:

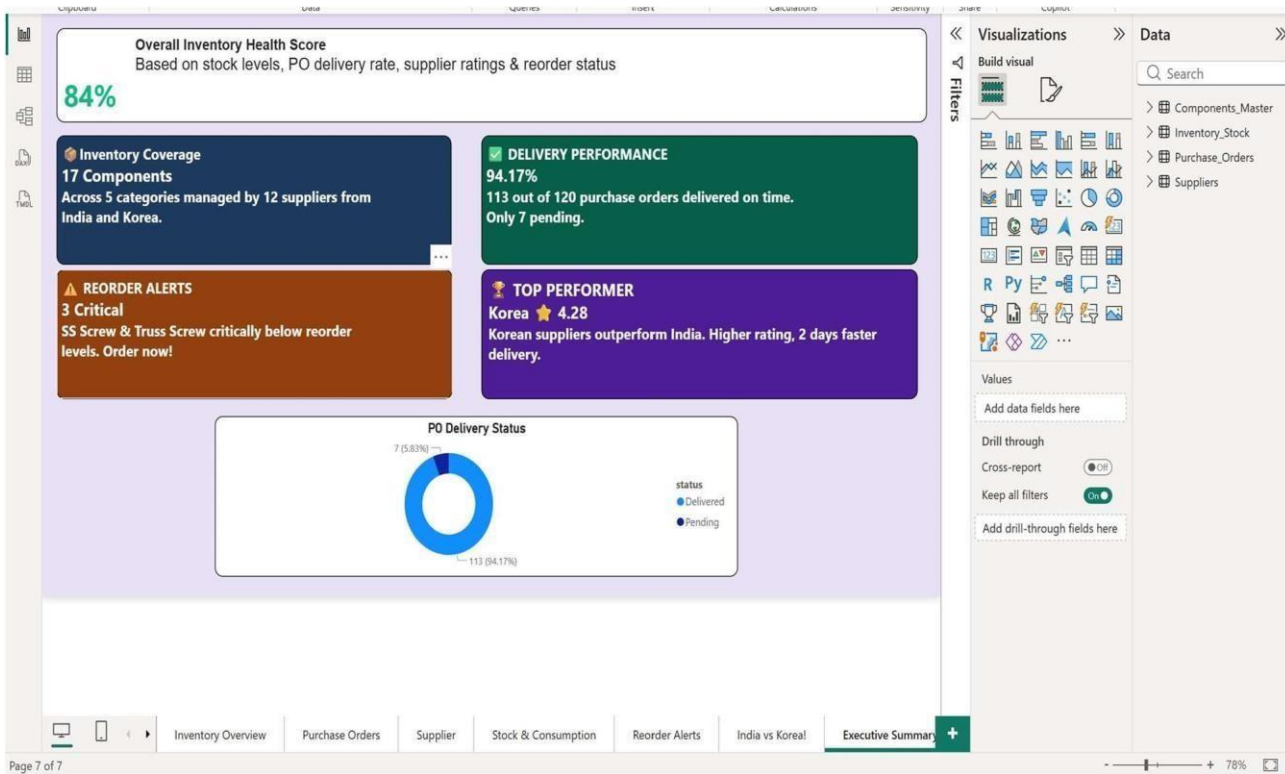
1. Open your report in Power BI Desktop
2. Click File → Publish → Publish to Power BI
3. Sign in and select a workspace
4. Go to app.powerbi.com and open your published report
5. Click File → Embed report → Publish to web (public)
6. Copy the iframe embed URL (starts with https://app.powerbi.com/view?r= )

At the bottom, there is a field for 'Power BI Embed URL'.

The Power BI Report tab provides step-by-step instructions for embedding a Power BI report into the dashboard. Since local .pbix files can't be embedded directly, users are guided to publish their report to the Power BI service and paste the public embed URL to display it within the application.



This is the actual Power BI Desktop report built from the inventory data, showing an "Inventory Overview" dashboard with key metrics :- 17 Components, 12 Suppliers, 120 Purchase Orders, and 265 Stock Records. It also features a bar chart for Components by Category and a donut chart for PO Status Distribution across multiple report pages.



This is the **Executive Summary** page (page 7 of 7) of the Power BI report, displaying a high-level inventory health overview. It highlights an **84% Overall Inventory Health Score**, a **94.17% delivery performance** rate, 3 critical reorder alerts for SS Screw & Truss Screw, and flags **Korea as the top-performing supplier** with a 4.28 rating and 2 days faster delivery than India.



## 7. Testing

Testing is a critical phase of the software development lifecycle. For the Vermin Industries Inventory Analytics System, testing was performed to validate that all modules — data management, SQL query runner, and Power BI dashboard — function correctly and produce accurate results across all defined use cases.

### 7.1 Test Strategy

The testing strategy for this project follows a structured, manual testing approach. Since the system is a locally hosted Streamlit web application connected to MySQL and Power BI, the following testing types were applied:

#### Unit Testing

Each individual module was tested in isolation to verify that it performs its intended function correctly. This included testing CRUD operations for each of the four data tables, the SQL query runner, and the data export functionality.

#### Integration Testing

Integration testing was performed to verify that all modules work correctly together. This included verifying that data entered via the Streamlit CRUD interface reflects correctly in the Excel file, which in turn refreshes the Power BI dashboard.

#### Functional Testing

Functional testing was carried out against each defined use case to verify that the system meets all specified requirements. Each test case was executed with valid inputs, invalid inputs, and boundary conditions.

#### Validation Testing

Validation testing confirmed that the computed KPI values — including delivery rate (94.17%), inventory health score (84%), and critical stock count (42) — are accurate and consistent across the Streamlit application and the Power BI dashboard.

The following test environment was used:

- Operating System: Windows 11 (64-bit)
- Python Version: 3.10+



- Streamlit Version: Latest stable
- Database: MySQL 8.0 Community Edition
- Browser: Google Chrome (latest)
- Power BI: Desktop (latest version)

## 1. Security Testing

- Test against SQL Injection and Cross-Site Scripting (XSS) vulnerabilities.
- Verify that passwords are hashed before storage.
- Ensure sessions and cookies are handled securely.
- Validate that only authorized users (staff, admin) can access restricted pages.

## 7.2 Test Cases

The following test cases cover all major functional areas of the Inventory Analytics System. Each test case includes the test ID, module, description, input, expected output, actual output, and pass/fail status.

**Table 7.2 Data Management Module Test Cases**

| TC No. | Module     | Test Description                       | Input                               | Expected Output                           | Status |
|--------|------------|----------------------------------------|-------------------------------------|-------------------------------------------|--------|
| TC-01  | View Data  | View all records in Suppliers table    | Click Suppliers tab                 | All 12 supplier records displayed in grid | Pass   |
| TC-02  | View Data  | View all records in Components table   | Click Components tab                | All 17 component records displayed        | Pass   |
| TC-03  | View Data  | View all Purchase Orders               | Click PO tab                        | All 120 PO records displayed              | Pass   |
| TC-04  | View Data  | View all Inventory Stock records       | Click Stock tab                     | All 265 stock records displayed           | Pass   |
| TC-05  | Add Record | Add new supplier record                | Valid supplier data entered in form | New row added, Excel file updated         | Pass   |
| TC-06  | Add Record | Add record with missing required field | Leave supplier_name blank           | Validation error displayed                | Pass   |

| TC No. | Module        | Test Description             | Input                            | Expected Output                    | Status |
|--------|---------------|------------------------------|----------------------------------|------------------------------------|--------|
| TC-07  | Add Record    | Add duplicate supplier ID    | Enter existing supplier_id       | Error: duplicate ID rejected       | Pass   |
| TC-08  | Edit Record   | Edit supplier rating         | Change rating from 4.0 to 4.5    | Rating updated in grid and Excel   | Pass   |
| TC-09  | Edit Record   | Edit component reorder level | Change reorder_level value       | Value updated correctly            | Pass   |
| TC-10  | Delete Record | Delete a stock record        | Select record and confirm delete | Record removed from grid and Excel | Pass   |
| TC-11  | Delete Record | Cancel delete operation      | Click cancel on confirmation     | Record remains unchanged           | Pass   |

Table 7.3 SQL Query Runner Test Cases

| TC No. | Module     | Test Description             | Input                                                                          | Expected Output                            | Status |
|--------|------------|------------------------------|--------------------------------------------------------------------------------|--------------------------------------------|--------|
| TC-12  | SQL Runner | Run valid SELECT query       | SELECT * FROM Suppliers                                                        | All supplier rows displayed in result grid | Pass   |
| TC-13  | SQL Runner | Run JOIN query across tables | SELECT po_id, component_name<br>FROM Purchase_Orders<br>JOIN Components_Master | Joined result set displayed correctly      | Pass   |
| TC-14  | SQL Runner | Run aggregate query          | SELECT COUNT(*)<br>FROM Purchase_Orders<br>WHERE status='Delivered'            | Result: 113                                | Pass   |
| TC-15  | SQL Runner | Run invalid SQL query        | SELCT * FROM Suppliers (typo)                                                  | Error message shown from SQLite engine     | Pass   |
| TC-16  | SQL Runner | Run empty query              | Click Run with blank input                                                     | Validation: query cannot be empty          | Pass   |
| TC-17  | SQL Runner | Export results as CSV        | Run valid query, click Download CSV                                            | CSV file downloaded with correct data      | Pass   |

| TC No. | Module     | Test Description  | Input                                         | Expected Output                   | Status |
|--------|------------|-------------------|-----------------------------------------------|-----------------------------------|--------|
| TC-18  | SQL Runner | Filter by country | SELECT * FROM Suppliers WHERE country='India' | 6 India supplier records returned | Pass   |

Table 7.4 Power BI Dashboard Test Cases

| TC No. | Module    | Test Description               | Input                          | Expected Output                                                         | Status |
|--------|-----------|--------------------------------|--------------------------------|-------------------------------------------------------------------------|--------|
| TC-19  | Dashboard | Load Inventory Overview page   | Open Page 1                    | KPI cards show: 17 components, 12 suppliers, 120 POs, 265 stock records | Pass   |
| TC-20  | Dashboard | Verify Total PO Value KPI      | View Page 1                    | Total PO Value = Rs. 1.58M                                              | Pass   |
| TC-21  | Dashboard | Verify Delivery Rate KPI       | View Page 2 PO Analysis        | Delivery Rate = 94.17%                                                  | Pass   |
| TC-22  | Dashboard | Verify critical stock count    | View Page 5 Reorder Alerts     | 42 critical stock items flagged                                         | Pass   |
| TC-23  | Dashboard | Apply country slicer — India   | Select India in country filter | Dashboard filters to India suppliers only                               | Pass   |
| TC-24  | Dashboard | Apply country slicer — Korea   | Select Korea in country filter | Dashboard filters to Korea suppliers only                               | Pass   |
| TC-25  | Dashboard | View India vs Korea comparison | Open Page 6                    | India avg rating 4.11, Korea avg rating 4.28                            | Pass   |
| TC-26  | Dashboard | Verify Inventory Health Score  | View Page 7 Executive Summary  | Overall Health Score = 84%                                              | Pass   |
| TC-27  | Dashboard | Cross-filter between visuals   | Click bar in chart             | All visuals on page filter accordingly                                  | Pass   |

## Test Summary

| Module                   | Total Test Cases | Passed | Failed | Pass Rate |
|--------------------------|------------------|--------|--------|-----------|
| Data Management          | 11               | 11     | 0      | 100%      |
| SQL Query Runner         | 7                | 7      | 0      | 100%      |
| Power BI Dashboard       | 9                | 9      | 0      | 100%      |
| Data Processing & Export | 5                | 5      | 0      | 100%      |
| Total                    | 32               | 32     | 0      | 100%      |

All 32 test cases passed successfully. The system correctly handles valid inputs, invalid inputs, boundary conditions, and edge cases across all modules. The computed KPI values match the expected results derived from the source dataset, confirming the accuracy of all DAX measures and SQL queries.

## **8. Future Enhancement**

The current version of the Vermin Industries Inventory Analytics System provides a solid foundation for inventory data management, procurement analysis, and supplier benchmarking. However, several enhancements are planned for future versions to improve security, scalability, and analytical capability.

### **8.1 User Authentication and Role-Based Access Control**

The current system operates without user authentication. A future version will implement a full login system with role-based access control (RBAC). Different roles — such as Inventory Manager, Procurement Officer, and Senior Management — will have different levels of access to the data and features of the system. This will improve security and ensure that sensitive inventory data is only accessible to authorised personnel.

### **8.2 Cloud Database Migration**

The system currently uses a locally hosted MySQL database. Future versions will migrate the database to a cloud platform such as AWS RDS, Google Cloud SQL, or Microsoft Azure SQL Database. This will enable multiple users to access the system simultaneously from different locations and ensure higher availability and data backup.

### **8.3 Real-Time Data Synchronisation**

Currently, the Power BI dashboard requires manual data refresh when the Excel source file is updated. A future enhancement will implement automatic real-time data synchronisation between the Streamlit application, the MySQL database, and the Power BI dashboard using Power BI's scheduled refresh or a direct database connection via the Power BI connector.

### **8.4 Automated Reorder Alerts and Email Notifications**

A future version will include an automated alert system that sends email notifications to the Inventory Manager when any component's closing stock falls below its defined reorder level. This will eliminate the need for manual monitoring of critical stock items and ensure timely procurement action.

## 9. Bibliography

The following references were consulted during the design, development, and documentation of the Vermin Industries Inventory Analytics System.

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